Number Representation: Decimal to Binary and Hexadecimal

Data Representation & Binary Number Systems • Standard: CCSS.MATH.PRACTICE.MP5

BOOK STORY OPENER: Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-Bit IBM 7090 Word

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using an 8-slot Ten-Frame register and [BASE], students decode bytes, hex nibbles, and bitwise masks.

Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 10: Computer Science, Base-N & 36-Bit Word Logic

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: 8-Bit Ten-Frame Register & Place Value Blocks

Use the 8 wells of the Ten-Frame as an 8-bit byte. Place pegs in wells for active '1' bits, leave empty for '0'.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Place pegs at positions 0, 1, and 3: binary 00001011_2 = 8 + 2 + 1 = 11_10. Predict hex value: 0B_16.

3. CALCULATOR ACTION (SC-32 KEYCAPS & STACK TRACE)

Keystroke Flow: [1] [1] [↖] [BASE]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [1]	`11`	Enter decimal integer 11 in DEC mode
[↖] [BASE]	`HEX 0B`	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↖] [BASE]	`BIN 1011`	Press [f:] [BASE]: screen displays Binary 1011!
[↖] [BASE]	`OCT 13`	Press [f:] [BASE]: screen displays Octal 13!
[↖] [BASE]	`DEC 11`	Press [f:] [BASE]: returns to Decimal 11

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Computer memory stores identical bit patterns; base conversions alter only the human visual representation.

Core Investigation Problem:

Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Bitwise Logic: AND Masking & Network Subnets

Boolean Logic & Bitwise Operations • Standard: CCSS.MATH.PRACTICE.MP5

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Row 1 holds byte A (1100 = 12). Row 2 holds mask B (1010 = 10). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[1] [2] [ENTER] [1] [0] [AND]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1] [2]	12	Enter operand A = 12 (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand B = 10 (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block Remediation: Model 12 (1100_2) and 10 (1010_2) on two rows of the Ten-Frame using pegs for 1s and empty wells for 0s; only columns where BOTH rows have a peg remain 1 in 12 AND 10 = 8 (1000_2).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block Remediation: Build 1 sixteen-rod and 0 ones on the Place Value tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

4. EXTENSION MATHEMATICAL INSIGHT

Bitwise AND masks bits, clearing unwanted bits to zero. This is the exact operation used in IPv4 subnet masking.

Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: NUMBER BASES & BITWISE LOGIC GATES

- Binary (Base 2): digits 0 and 1. Hexadecimal (Base 16): digits 0-9 and A-F.
- Press [f:] [BASE] to cycle between DEC (decimal), HEX (hexadecimal), OCT (octal), and BIN (binary).
- Bitwise AND: result bit is 1 only if BOTH bits are 1 (used for masking and subnets).

Key Sequence: 1 1 ↖ BASE

Problem 1: Convert decimal 42 to binary using [f:] [BASE].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 2: Convert hexadecimal 2A to decimal.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Unit 12: Digital Logic, Base Conversions & Computer Architecture — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–12 / AP CS

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Core Solution: HEX: FF, BIN: 11111111 (8 ones). Keystrokes: [2][5][5] [f:][BASE]

Core Check Keys: `2` `5` `5` `↖` `BASE`

Extension Challenge: Calculate bitwise OR for 12 (1100) and 3 (0011). Predict the binary and decimal result.

Extension Solution: Binary: 1111, Decimal: 15. Keystrokes: [1][2] [ENTER] [3] [OR]

Ext. Check Keys: `1` `2` `ENTER` `3` `OR`

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Convert decimal 42 to binary using [f:] [BASE].

Expected Answer: 101010 (32 + 8 + 2)

SC-32 Verification Keys: `4` `2` `↖` `BASE`

Problem 2: Convert hexadecimal 2A to decimal.

Expected Answer: 42

SC-32 Verification Keys: `In` `HEX` `mode:` `2` `A` `->` `switch` `to` `DEC`

Problem 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

Expected Answer: 7 (0111)

SC-32 Verification Keys: `1` `5` `ENTER` `7` `AND`

Problem 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

Expected Answer: 6 (0110)

SC-32 Verification Keys: `1` `2` `ENTER` `1` `0` `XOR`

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.